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Heat Transfer Enhancement in Heat Exchanger Using Porcupine Fins and Al2O3-Water Nanofluid

Type: Bachelor's project in Mechanical Engineering
Institution: Government Engineering College, Kozhikode
University: APJ Abdul Kalam Technological University

Software / methods used:
- ANSYS DesignModeler
- ANSYS Fluent
- CFD simulation of a double-pipe heat exchanger
- Comparative analysis using porcupine fins and Al2O3-water nanofluid

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Declaration, certificate, and signature pages were removed.
This was completed as a group academic project.

Prepared by

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ABSTRACT

This study investigates the use of porcupine fins to enhance heat transfer in a heat exchanger. Porcupine fins are a novel type of heat transfer surface that have been shown to increase heat transfer rates in various applications. In this study, a heat exchanger with porcupine fins is designed and evaluated through experimental testing under various operating conditions. The results indicate that the use of porcupine fins yields significant improvements in heat transfer performance compared to conventional heat exchangers. The findings suggest that the use of porcupine fins can be an effective strategy for improving the efficiency of heat exchangers in various industrial applications. The results of this study have important implications for the design and optimization of heat exchangers for thermal management in various fields such as aerospace, automotive, and energy industries. The nanofluid, consisting of Al_20_3 nanoparticles dispersed in water, is used as the heat transfer medium to enhance the thermal conductivity of the fluid. The performance of the heat exchanger is evaluated through experimental testing under various operating conditions. The results indicate that the combination of porcupine fins and Al_20_3-water nanofluid yields significant improvements in heat transfer performance compared to conventional heat exchangers. The findings suggest that the use of porcupine fins and nanofluids can be an effective strategy for improving the efficiency of heat exchangers in various industrial applications.

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ABBREVIATIONS

1. FVM - finite volume method
2. CFD - Computational Fluid Dynamics
3. CP-SP - Constant physical properties single-phase
4. VP- SP - Variable physical properties single-phase

CHAPTER 1
INTRODUCTION

Heat exchangers are essential components in various industrial processes, such as power generation, chemical production, and refrigeration systems. The performance of these heat exchangers is highly dependent on their ability to transfer heat efficiently. However, the heat transfer efficiency of conventional heat exchangers can be limited due to factors such as low thermal conductivity of the fluid, low heat transfer coefficient, and fouling.

Finned surfaces and nanofluids are two contemporary approaches that have been developed to improve the heat transfer performance of heat exchangers. In this study, we look into how porcupine fins and an Al_20_3-water nanofluid can improve heat transmission in a heat exchanger.

Porcupine fins are a kind of surface enhancing method that uses tiny, spiky fins that stick out from the heat transfer surface. The fluid flow is disturbed by these fins, which raises the convective heat transfer coefficient. The heat transfer fluid known as a "nanofluid" has nanoparticles suspended in a base fluid, on the other hand. These nanoparticles can considerably raise the fluid's thermal conductivity, which improves heat transfer efficiency.

In this project, we carried out numerical simulation to find out how porcupine fins and an Al_20_3-water nanofluid affect a heat exchanger's ability to transfer heat. Under various working conditions, we examined the heat transfer coefficient, pressure drop, and overall heat transfer rate.

The findings of this study can have significant implications for the design and optimization of heat exchangers in various industrial applications. For example, the findings of the study can be used to select the most appropriate type of fluid for a particular application, to determine the optimal flow rate, and to design a heat exchanger that can achieve the desired heat transfer performance. This report presents a detailed description of our numerical simulation, methodology, and results, followed by a discussion of the implications and potential future directions for this research.

1.1. FINS

Fins are thin protrusions or extensions that are affixed to a heat transfer device's surface. Depending on the application and the type of heat transfer issue, fin size and shape might vary greatly. The three shapes of fins that are most frequently employed are round, triangular, and rectangular. Depending on the operating circumstances and the desired qualities, fins can be constructed from a variety of materials, including metals, polymers, ceramics, and composites. A fundamental mechanism that is essential in many engineering applications is heat transfer through fins. The method involves transmitting heat through a finned surface, which greatly accelerates the rate of heat transfer, from a surface to the fluid around it. Fins are often used in many thermal systems, including air-cooled heat exchangers. By adding fins, a heat transfer device can increase its surface area and hence increase the efficiency of its heat transfer. The principles of conduction, convection, and radiation serve as the foundation for the mechanism of heat transmission through fins. The fin conducts the heat from the base surface, which is subsequently transferred to the surrounding fluid, usually air. The rate of heat transmission through a finned surface is influenced by a number of variables, including the fluid's characteristics, the temperature difference between the surface and the surrounding fluid, and the fin geometry and material. Therefore, a detailed understanding of these elements and how they interact is necessary to design an ideal finned surface.

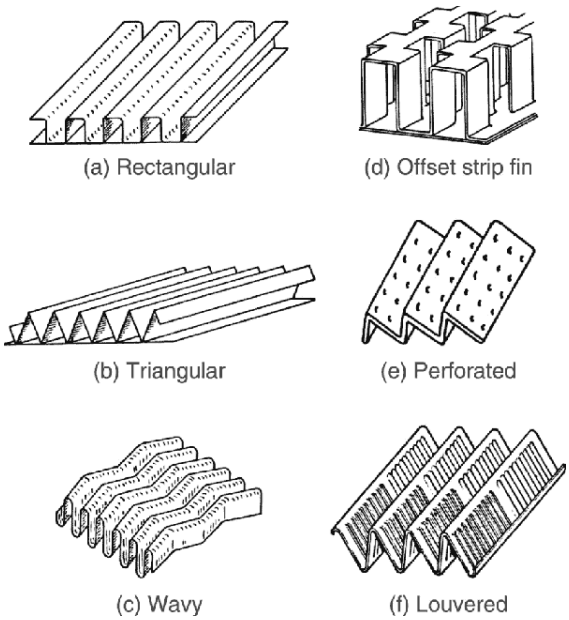


Fig 1.1 Types of fins [1]

1.2 PORCUPINE FINS

Porcupine fins are a type of heat exchanger fin design that resembles the quills of a porcupine. These fins have a series of small, sharp, and pointed projections that extend outward from the base of the fin. The purpose of these projections is to increase the surface area of the fin and improve its heat transfer efficiency. Porcupine fins consist of a series of sharp, pointed metal spikes that are arranged in a circular or helical pattern around the surface of the heat transfer tube or plate.

The spikes are typically angled in a way that directs the fluid flow over them, which increases the rate of heat transfer between the fluid and the heat transfer surface. It's important to note that the design of porcupine fins must be carefully optimized to achieve the best performance in each specific heat exchanger application. Therefore, it's recommended to consult with a qualified engineer or heat transfer specialist to determine the most appropriate design for your specific needs.

The dimensions of porcupine fins can vary depending on the specific application and design requirements. Generally, the height and thickness of the fins can range from a few millimetres to several centimetres, while the distance between the fins can range from a fraction of a millimetre to a few millimetres. Porcupine fins can be used in various types of heat exchangers, including plate-fin heat exchangers and air-cooled heat exchangers.

The design of porcupine fins provides several advantages over other types of fins. Because the fins are sharp and pointed, they create a turbulent flow in the fluid passing over them, which enhances heat transfer by breaking up the fluid boundary layer and increasing the surface area of the heat transfer surface. The sharp edges also minimize the accumulation of dirt and debris on the fins, which can reduce the efficiency of other types of fin designs.

Additionally, the high surface area of the porcupine fins means that they can be used in compact heat exchanger designs, making them well-suited for applications where space is limited. However, the sharp edges of porcupine fins can also increase the pressure drop across the heat exchanger, which can be a disadvantage in some applications.

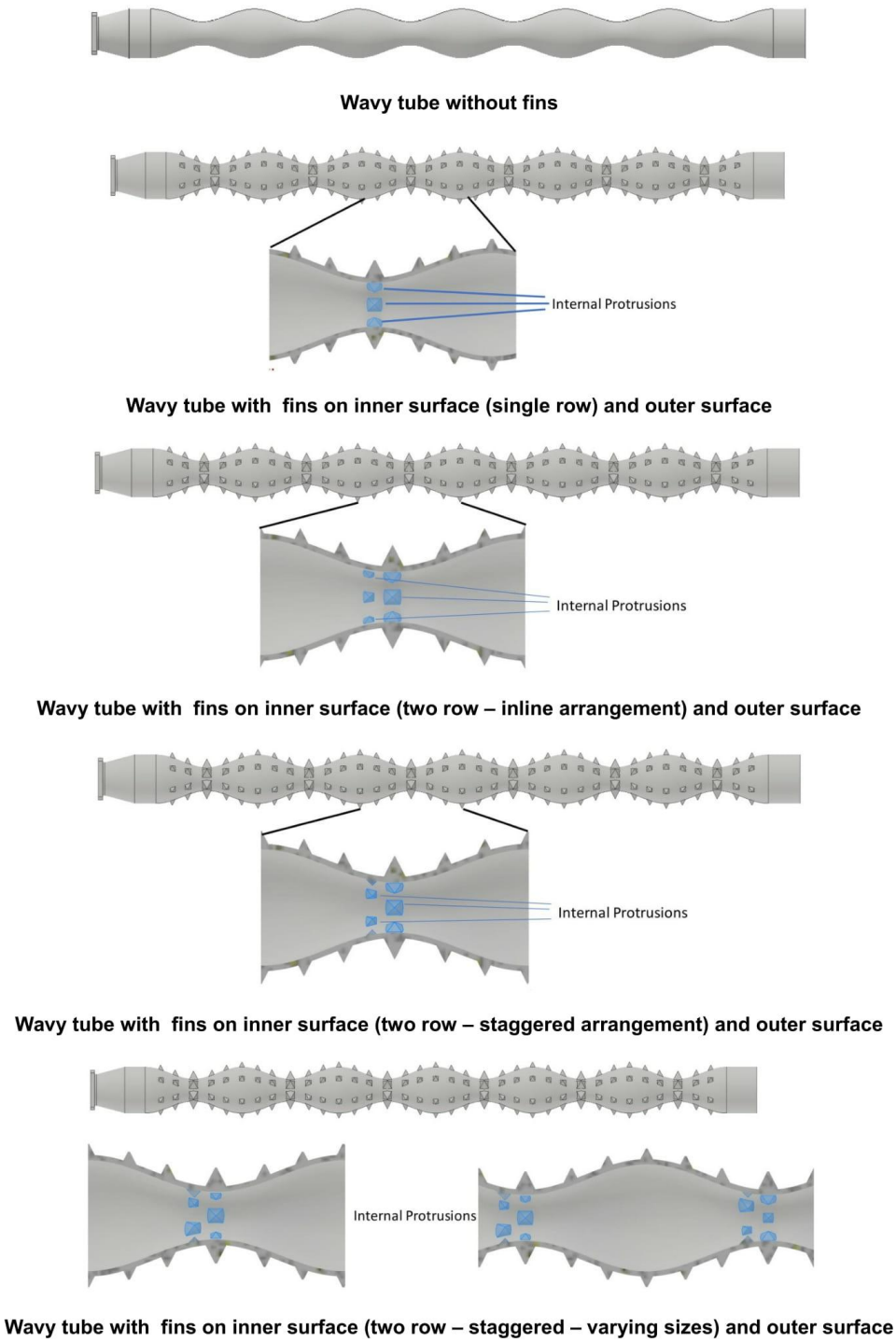
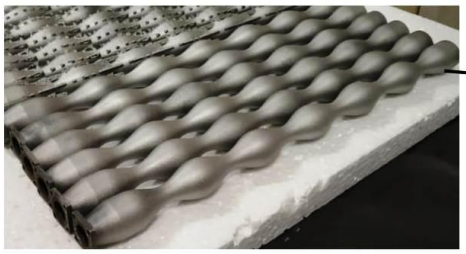


Fig 1.2. Graphical representation of porcupine fins [2]

First set of 3D printed models (different configurations)



Plain wavy tubes

Plain tubes (for comparison)

Porcupine fin on the outer surface with varying fin sizes

All the models area manufactured by 3D metal printing technology

Fig 1.3. Real models of porcupine fins [2]

1.3 NANOFLUIDS

Fluids called nanofluids have nanoparticles suspended in them. These nanoparticles often fall within the nanoscale range, meaning that their sizes fall between 1 and 100 nanometers. A base fluid, which is frequently a common liquid like water, oil, or ethylene glycol, is used to disseminate the nanoparticles.

The addition of nanoparticles to the base fluid alters its properties, resulting in a nanofluid with enhanced thermal, electrical, or optical characteristics. The choice of nanoparticles depends on the desired properties and application of the nanofluid. Common nanoparticles used in nanofluids include metals (such as copper, silver, or gold), metal oxides (such as titanium dioxide or alumina), or carbon-based materials (such as carbon nanotubes or graphene).

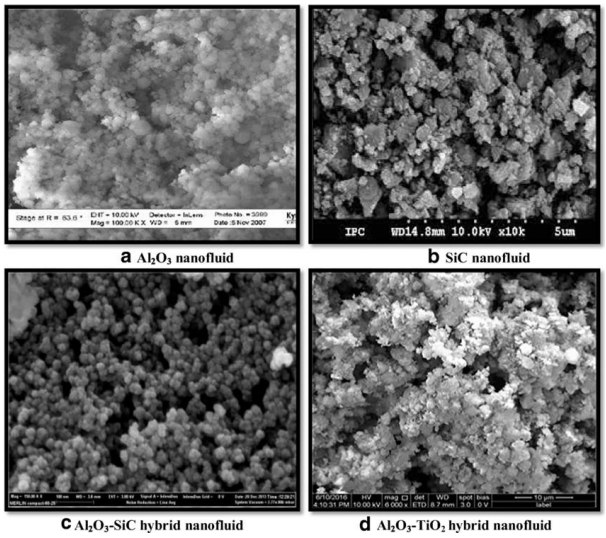


Fig 1.4 SEM images of 0.1% wt. fraction of (a) Al_2O_3 nanofluid (b) SiC nanofluid, (c) Al_2O_3-SiC hybrid nanofluid, (d) Al_2O_3-TiO_2 hybrid nanofluid, [3]

Nanofluids have gained significant attention due to their unique properties. For example, they exhibit enhanced thermal conductivity compared to the base fluid, which makes them suitable for applications in heat transfer and cooling systems. Additionally, nanofluids can exhibit improved optical properties, electrical conductivity, and stability.

CHAPTER 2
LITERATURE REVIEW

Hooman Yarmand et al. [1] (2014) conducted numerical investigation of heat transfer enhancement through a rectangular heated pipe for turbulent nanofluid. The finite volume method (FVM) was applied to solve the continuity, momentum, and energy equations. The top and bottom of the symmetrical rectangular channel provide a constant heat flux while the sides are insulated. Various volume fractions of nanofluids, including Al_2O_3, ZnO, CuO, and SiO_2 nanoparticles of four different types, are taken into consideration in the current experiment. The numerical results demonstrate that, when compared to other nanofluids, SiO_2-water has the highest Nusselt number and the lowest heat transfer coefficient due to its low thermal conductivity. Along with the Nusselt number, both the Reynolds number and the volume fraction of nanoparticles increase.

A Azari et al. [2] (2014) conducted CFD and experimental investigation on the heat transfer characteristics of alumina nanofluids under the laminar flow regime. In this study, computational fluid dynamics (CFD) and experimental techniques are used to evaluate the laminar convective heat transfer coefficient of Al_2O_3/water nanofluids in a circular tube under a uniform and continuous heat flow on the wall. A model with changing physical features in one phase, a model with stable physical qualities in one phase, and a discrete particles two-phase model (CP-SP) were all developed. Particle aggregation and cluster size dispersion were both considered in the two-phase model. Experimental and modeling+ investigations have shown that nanofluids operate thermally better than the base fluid, and that the improvement in heat transfer increases with particle volume concentration and Reynolds number.

Faris Mohammed Ali et al. [3] (2011) performed study of the effect of volume fraction concentration and particle materials on thermal conductivity and thermal diffusivity of nanofluids. In this study, we give fresh information regarding the improvement of thermal conductivity in four nanofluids including distilled water and aluminium oxide (Al_2O_3) nanoparticles with diameters of 11, 25, 50, and 63 nm. The base fluid was collected in an ultrasonic device for roughly 7 hours while the nanofluids were created utilizing a single step

process (directly dispersing the nanoparticle in the base fluid). The produced nanofluids' thermal diffusivity and conductivity were measured using the transient hot-wire laser beam displacement method. By adjusting the experimental results to the numerical data generated by simulating Al_2O_3 in distilled water, the thermal conductivity and thermal diffusivity were determined. The conclusion reached was that when particle size grows, nanofluids' thermal conductivity and thermal diffusivity enhancement rise.

Yuan Xue [4] was made to enhance the heat transfer of plate fin heat exchanger further, three new kinds of wavy plate fins, namely perforated wavy fin, staggered wavy fin and discontinuous wavy fin, are proposed and investigated by computational fluid dynamics (CFD) simulations. With Reynolds numbers varying from 500 to 4500, the effects of important design elements such as waviness aspect ratios, perforation sizes, staggered ratios, and breaking distance are examined. The perforated wavy fin has a greater Nusselt number and friction factor than the conventional wavy fin. In comparison to the conventional wavy fin, the serration technique is advantageous in reducing the friction factor. It is apparent that the serration approach in the reduced aspect ratio can enhance the wavy fin's thermal-hydraulic performance.

Mao-Yu Wen et al[5] made three different fins (plate fin, wavy fin, and compounded fin) and were investigated in a wind tunnelThe discussion has included the heat transfer coefficient, the air side pressure drop, the Colburn factor (j), and the fanning friction factor (f) in relation to air velocity (1-3 m/s) and Reynolds number (600-2000). Flow visualization was developed to monitor the specific fluid flow properties and offer insight on the fluid flow phenomena. Results reveal that the pressure drop, heat transfer coefficient, f factor, and j factor increase by around 10.9–31.9%, 11.8–24.0%, 2.2–27.5%, and 0.5-2.7%, respectively, when the wavy fin is compared to the flat fin. Additionally, the pressure drop, heat transfer coefficient, f factor, and j factor rise by roughly 33.5–63.1%, 27.0–45.5%, 6.9–71.1%, and 9.4–13.2%, respectively, when the compounded fin is compared to the flat fin.

Kabeel et al[6] studied the effect of aluminum and copper fins on heat transfer enhancement in a shell-and-tube heat exchanger . They found that copper fins significantly improved the heat transfer rate compared to aluminum fins. The authors attributed this to the higher thermal conductivity of copper.

Sravana Keerthi et al [7]. Conducted an experiment comparing the heat transfer performance of a work type heat exchanger with Al_2O_3 nanofluid to a conventional heat exchanger. The results showed that using nanofluids, particularly Al_2O_3-water nanofluid, in mesh-type and triangular fin-type heat exchangers significantly improved heat transfer efficiency. In counter flow configurations, triangular fin-type heat exchangers with nanofluids achieved an effectiveness of 0.76, indicating a substantial increase compared to other designs. Nanofluids offer a promising approach to enhance heat transfer rates in various heat exchanger configurations.

S Basavarajappa et al [8] compiled various research works carried out to improve the thermal performance of a heat exchanger by using different parameters. These parameters include the type of fins, their orientation, shapes, and locations. The purpose of the review is to understand how these parameters influence the improvement of thermal performance. The review likely covers a range of experiments and studies conducted by different researchers, each focusing on specific aspects of finned tube heat exchangers, such as comparing heat transfer enhancement and pressure drop in different types of heat exchangers, evaluating the effects of various fin geometries, investigating heat transfer rates, pressure drop measurements, Nusselt number, friction factor, and Rayleigh number.

Vinous M. Hameed et al [9] did an experimental investigation focused on studying the temperature distribution in a double pipe heat exchanger. A copper tube with a Perspex shell was used to construct the heat exchanger. The test section consisted of a horizontal heat exchanger with specific dimensions, insulation, and thermal paste application.

Water was used as the working fluid for the experiment, with hot water flowing through the inner tube and cold water through the shell side. To gauge the temperature distribution along the inner tube and at the intake and exit of both sides of the heat exchanger, eight K-type thermocouples were positioned at various points. A double pipe heat exchanger's temperature distribution and fluid temperature change were the subject of the investigation. By establishing optimal lengths based on temperature variations along the exchanger, the study's findings may be used to construct heat exchangers.

CHAPTER 3
OBJECTIVES AND METHODOLOGY

3.1 OBJECTIVES

- To conduct Computational Fluid Dynamics (CFD) analysis for double pipe heat exchanger equipped with porcupine fins.
- To study the influence of porcupine fins inserted on heat exchanger on Nusselt number and friction factor for a heat exchanger in turbulent flow regime.
- To compare the thermal enhancement factor for porcupine fins with that of normal double pipe heat exchanger with wavy inner tube and without fins.
- To find thermo physical properties of Al_20_3-water nanofluid numerically.
- Also compare the same condition with double pipe heat exchanger with porcupine fins, when Al_20_3-water nanofluid is used as fluid in outer tube.
- To study the results obtained and explain the physical reasons for the variation in Nusselt number, friction factor and the thermal enhancement factor with Reynolds number for the newly introduced Porcupine fins.

3.2 METHODOLOGY

- The fluid flow is analyzed using three equations. The three equations states about conservation of mass, conservation of momentum and conservation of Energy. They are Continuity equation, Navier-Stokes equation and Energy equations.
- The simulation is performed on ANSYS 2019 and all the above equations are solved using Ansys fluent
- Assign material properties to the heat exchanger components, including the porcupine fins and the fluid. The properties can include thermal conductivity, density, specific heat, and viscosity.
- Set up the fluid flow simulation in ANSYS Fluent. Define the fluid flow domains, boundary conditions, and initial conditions.
- Set up the heat transfer simulation by specifying the heat generation rate in the heat exchanger, and the heat transfer coefficient between the fluid and the fins.

- Run the simulation and monitor the convergence of the solution. Adjust the simulation settings as necessary to ensure convergence.
- Analyze the simulation results, including temperature distribution, pressure drop, and heat transfer rate. Compare the results of the simulation with and without porcupine fin and along with heat exchanger with porcupine fins (Al_20_3-water nanofluid)

CHAPTER 4
NUMERICAL SIMULATION

4.1 GEOMETRY

Modeling is the first step to numerical analysis like. . Here are some of the figures of 3D models which are created by using ANSYS 2019. Following models were made using design modeler in ANSYS 2019.

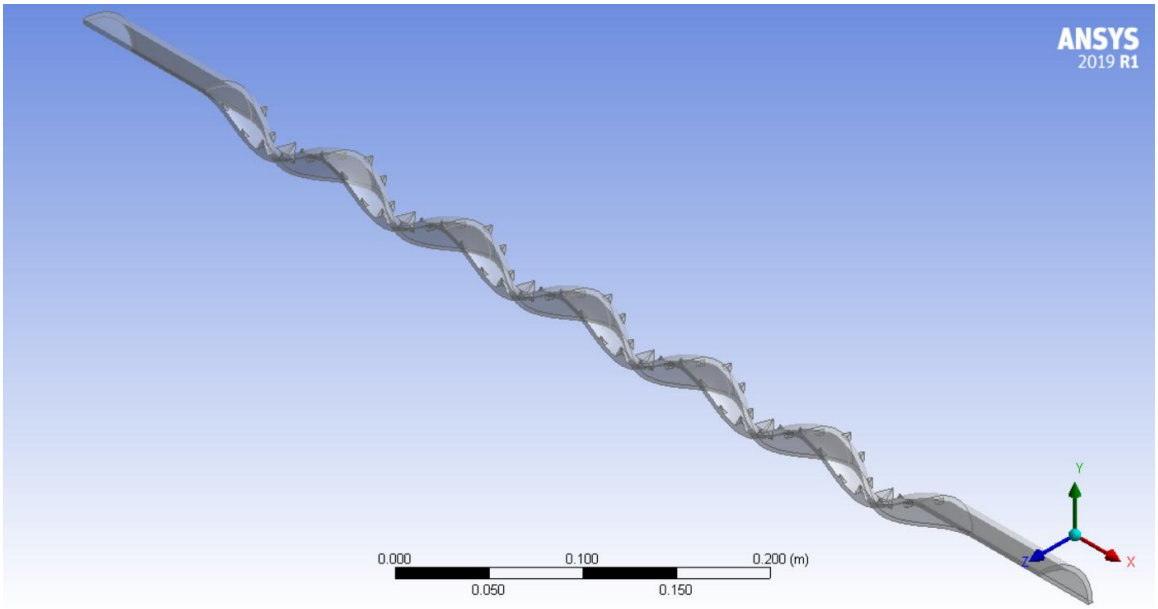


Fig 5.1 Porcupine fins on the outer surface of inner pipe

Parameter	symbol	Value
Length of single section of fin	L_1	90mm
Thichness of the section of fin	B_1	40mm
Distance between each protrusions of fin	D_1	15mm
Smaller protrusion height	h_1	5.45mm
Large protrusion height	H_1	32.03mm

Table 4.1 Specifications of porcupine fins

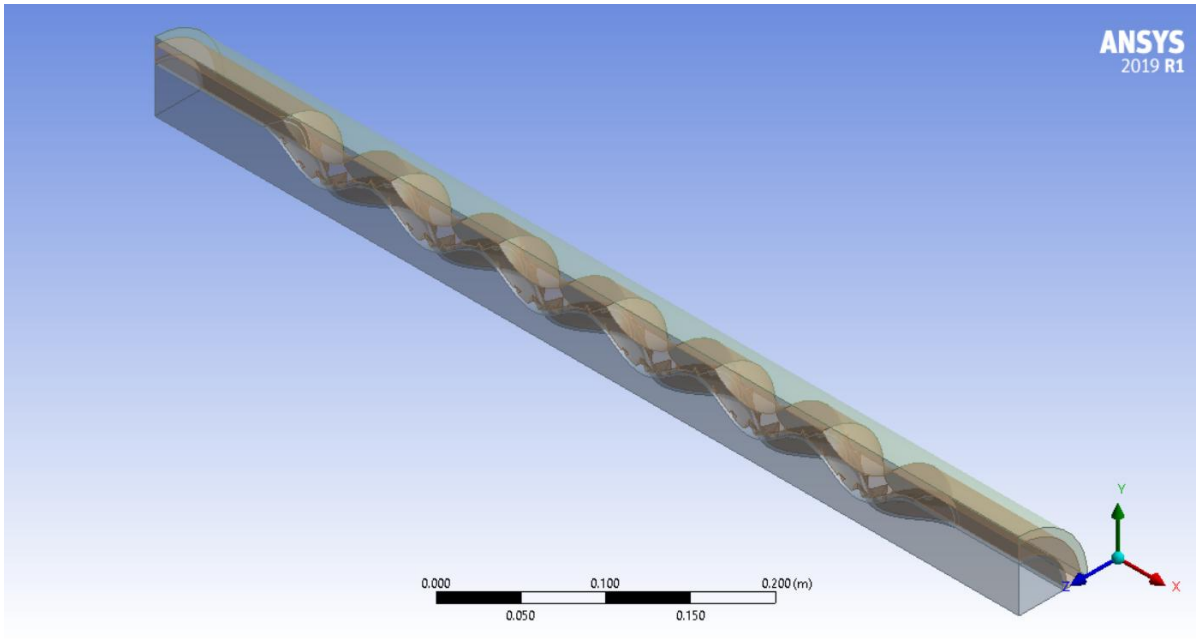


Fig 4.2 Double pipe heat exchanger with porcupine fins

Parameter	Symbol	Value
Length of heat exchanger	L_2	720mm
Thickness of heat exchanger	B	57.15mm
Length of portion of pipe with fins	L_3	540mm
Left portion of heat exchanger	L_4	90mm
Right portion of heat exchanger	L_5	90mm
Diameter of inner pipe	D_2	102.26mm
Diameter of outer pipe	D_3	114.3mm

Table 4.2 specification of double pipe heat exchanger

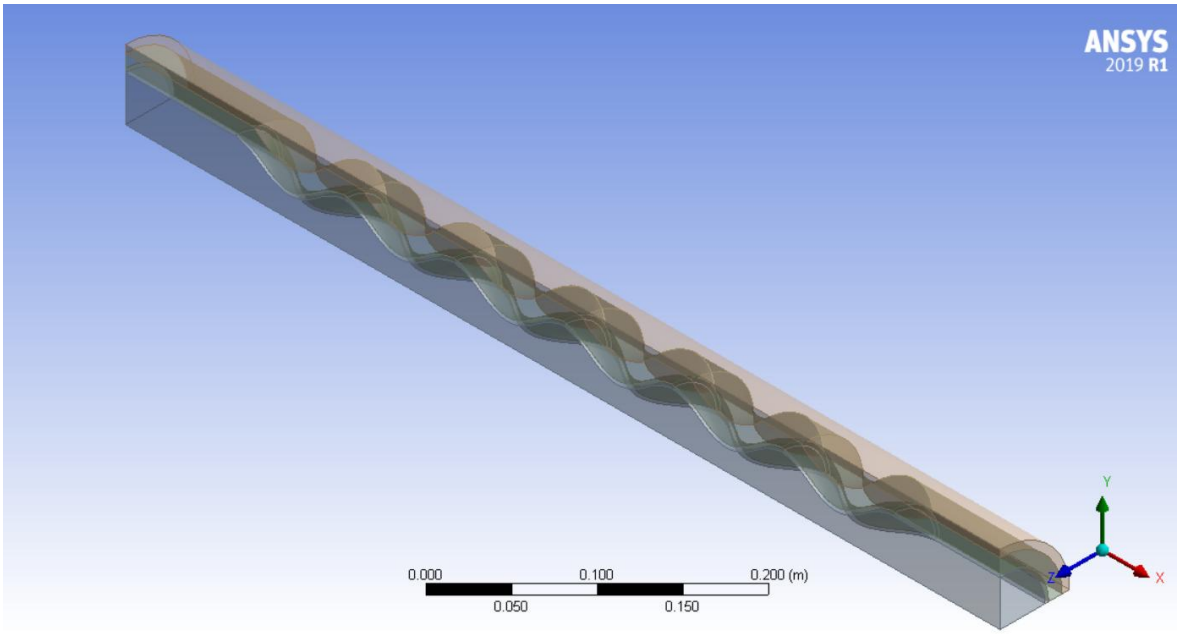


Fig 4.3 Double pipe heat exchanger without fins

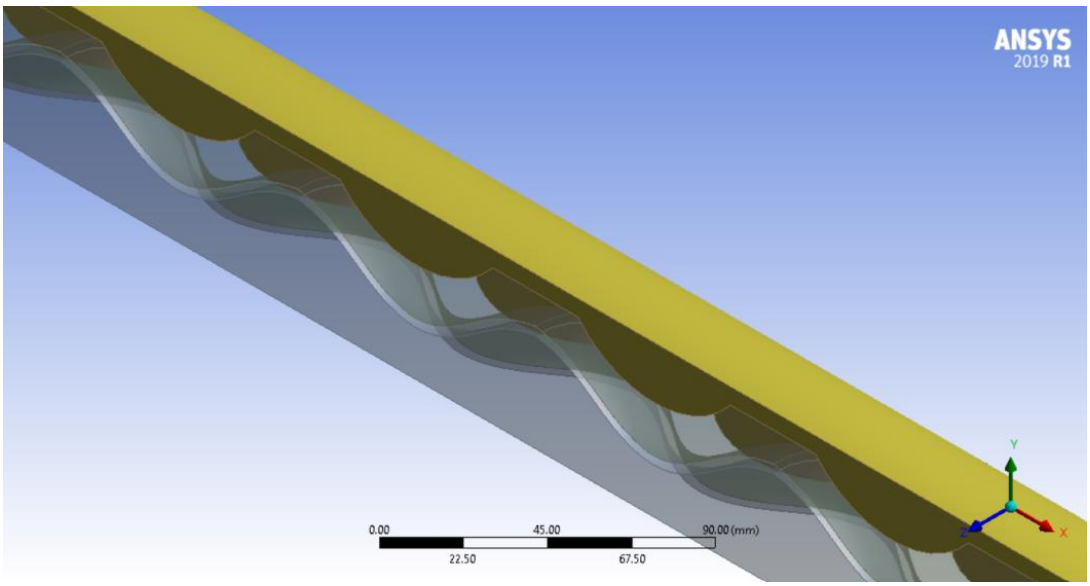


Fig 4.4 U- shaped structure on inner surface of outer pipe

4.2 MESH

The heat exchanger with porcupine fins underwent a grid independence test at a Reynolds number of 5000. The mesh's element count was changed from 715659 to 5638068 in order to study how the Nusselt number and friction factor changed over time. Figure illustrates the test's outcomes. The Nusselt number and friction factor were not significantly different from the 3007254 cells mesh as the trial progressed. The 3007254 cells mesh is fixed as the mesh that will be utilized for Nusselt number and friction factor predictions at all Reynolds numbers based on the observation of the mesh independence test results.

Figure 4.5 shows the mesh used for the simulation work. Inflation layer is provided near the wall so as to capture the effects of the boundary layer at the walls. The mesh used for the work consists of 3007254 elements.

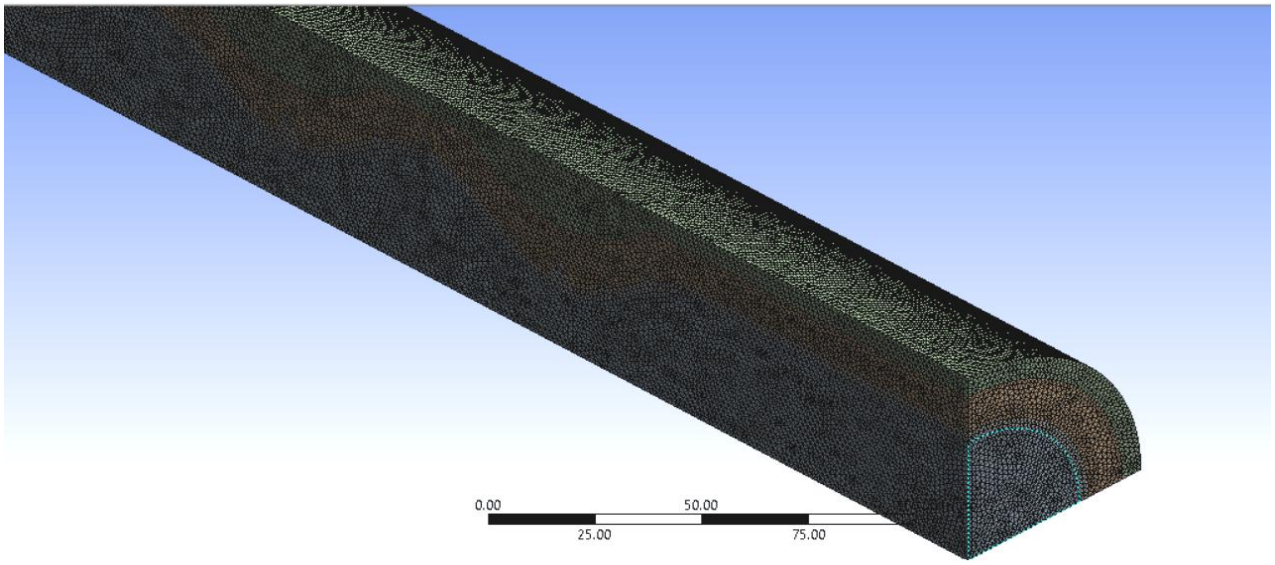


Fig 4.5 Mesh of the geometry

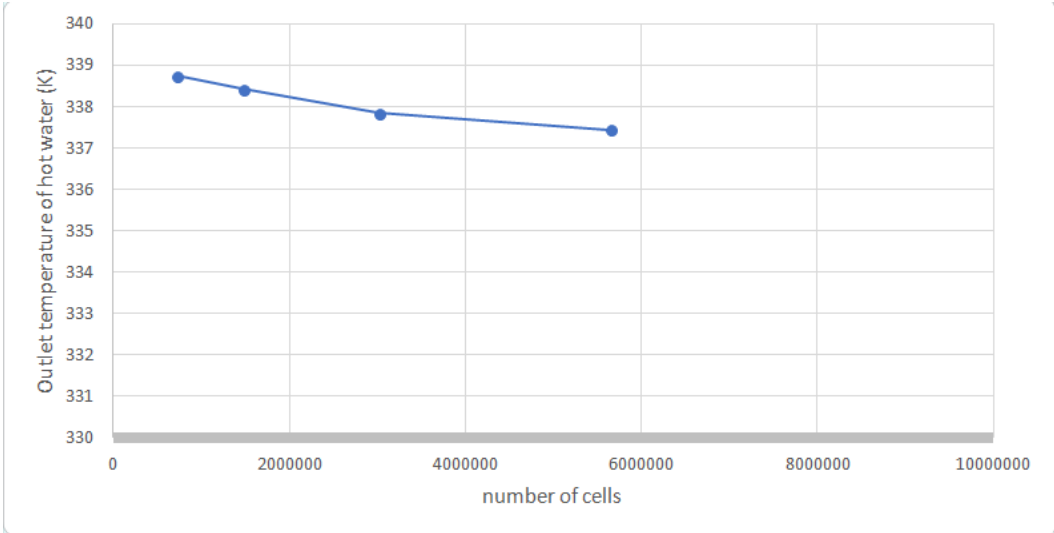


Fig 4.6 mesh independence test

4.3 BOUNDARY CONDITIONS

Pressure-based solver is used because the problem is concerned with incompressible fluid flow. The realizable $k - \varepsilon$ turbulence model is used with standard wall function. The outer wall of the outer pipe surface is made adiabatic. The material properties assigned for the heat exchanger are shown in table

Properties	Inner pipe	Outer pipe
Material	Copper	Aluminium
Thermal conductivity(W/mk)	387.6	202.4
Density (kg/m^3)	8978	2719
Specific heat (J/Kg K)	381	871

Table 4.3 material properties

In this work, hot water is made to flow through the inner pipe and cold water through the outer annulus. The inlet temperatures assigned for hot and cold fluid domains are 75^{0}C and 40^{0}C respectively. The Reynolds number for cold fluid domain is kept constant at 5000. A Reynolds number range of 5000-7000 with 500 increments is used for the hot fluid domain. The pressure-velocity coupled algorithm is used for the simulation. The convergence of solution was completed after approximately 1 hour 30 minutes for each Reynolds number.

The generalized equations that describe the fluid characteristics are as below:

Continuity equation

The continuity equation is a fundamental equation used to describe the conservation of mass in fluid flow. It states that the rate of mass entering or leaving a control volume is equal to the

rate of change of mass within that volume. Mathematically, the continuity equation can be expressed as:

$$\frac{1}{r}\frac{\partial (ru_r)}{\partial r} + \frac{1}{r}\frac{\partial u_\theta}{\partial \theta} + \frac{\partial u_z}{\partial z} = 0$$

Navier Stokes equation,

This equation is law of conservation of momentum. For a fluid flow there are four forces acting on a fluid element, they are, Inertia force, Viscous force, Body forces like gravity or magnetic forces, and pressure force. The Navier Stokes equations in cylindrical coordinates are given below,

$$\rho\left(\frac{\partial u_r}{\partial t} + u_r\frac{\partial u_r}{\partial r} + \frac{u_\theta}{r}\frac{\partial u_r}{\partial \theta} + u_z\frac{\partial u_r}{\partial z} - \frac{u_\theta^2}{r}\right)$$
$$= \rho g_r - \frac{\partial P}{\partial r} + \mu\left(\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u_r}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 u_r}{\partial \theta^2} + \frac{\partial^2 u_r}{\partial z^2} - \frac{2}{r^2}\frac{\partial u_\theta}{\partial \theta} - \frac{u_r}{r^2}\right)$$

$$\rho\left(\frac{\partial u_\theta}{\partial t} + u_r\frac{\partial u_\theta}{\partial r} + \frac{u_\theta}{r}\frac{\partial u_\theta}{\partial \theta} + u_z\frac{\partial u_\theta}{\partial z} - \frac{u_r u_\theta}{r}\right)$$
$$= \rho g_\theta - \frac{1}{r}\frac{\partial P}{\partial \theta} + \mu\left(\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u_\theta}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 u_\theta}{\partial \theta^2} + \frac{\partial^2 u_\theta}{\partial z^2} + \frac{2}{r^2}\frac{\partial u_r}{\partial \theta} - \frac{u_\theta}{r^2}\right)$$

$$\rho\left(\frac{\partial u_z}{\partial t} + u_r\frac{\partial u_z}{\partial r} + \frac{u_\theta}{r}\frac{\partial u_z}{\partial \theta} + u_z\frac{\partial u_z}{\partial z}\right)$$
$$= \rho g_z - \frac{\partial P}{\partial z} + \mu\left(\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u_z}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 u_z}{\partial \theta^2} + \frac{\partial^2 u_z}{\partial z^2}\right)$$

Energy equation,

This equation tells us about law of conservation of energy. The energy equation in cylindrical coordinate is given by,

$$\frac{\partial T}{\partial t} + u_r\frac{\partial T}{\partial r} + \frac{u_\theta}{r}\frac{\partial T}{\partial \theta} + u_z\frac{\partial T}{\partial z} = \frac{q_g^{\cdot}}{C_p} + \alpha\left(\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial T}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 T}{\partial \theta^2} + \frac{\partial^2 T}{\partial z^2}\right)$$

4.4 THERMO PHYSICAL PROPERTIES OF NANOFLUID

- Density of nanofluid

$$\rho_{nf} = (1 - \emptyset)\rho_{bf} + \emptyset\rho_{np}$$

- Specific heat of nanofluid

$$C_{p,nf} = \frac{(1 - \emptyset)\rho_{bf} C_{p,bf} + \emptyset\rho_{np} C_{p,np}}{\rho_{nf}}$$

- Thermal conductivity of nanofluid

$$k_{nf} = (1 - \emptyset)k_{bf} + \emptyset k_{np}$$

- Viscosity of nanofluid

$$\mu_{nf} = (\mu_{bf})(\frac{1}{(1 - \emptyset)^{2.5}})$$

Where ρ_{nf} is the density of nanofluid, ρ_{np} is density of nanoparticle, $\emptyset$ is volume fraction of nanoparticles, ρ_{bf} is the density of base fluid (water), $C_{p,bf}$ is the specific heat of base fluid, $C_{p,np}$ is the specific heat of nanoparticle, k_{bf} is the thermal conductivity of base fluid, k_{nf} is the thermal conductivity of nanoparticle, μ_{bf} is the viscosity of base fluid and μ_{nf} is the viscosity of nanofluid.

Fluid	Density (kg/m^3)	Specific heat (J/kg K)	Thermal conductivity (W/Mk)	Viscosity (kg/ms)
Al_2O_3-water nanofluid	1012.798	4105.3087	0.797	0.00101423

Table 4.4 Thermo physical properties of nanofluid

CHAPTER 5

VALIDATION

Experimental work was done on a double pipe heat exchanger with a counter current flow type. The inner tube and shell of the experimental test rig are composed of copper and Perspex, respectively. In order to comprehend the temperature distribution in a heat exchanger, Vinous M. Hameed and Sarah M. Awad's aforementioned experimental work was simulated by us. This study will help in the construction of an adequate heat exchanger length based on the value of the temperature change along the studied heat exchanger. So, as shown in figure 5.1, the relationship between axial distance and temperature distribution at the heat exchanger's tube side is discovered to be of the same trend and value once simulation is completed.

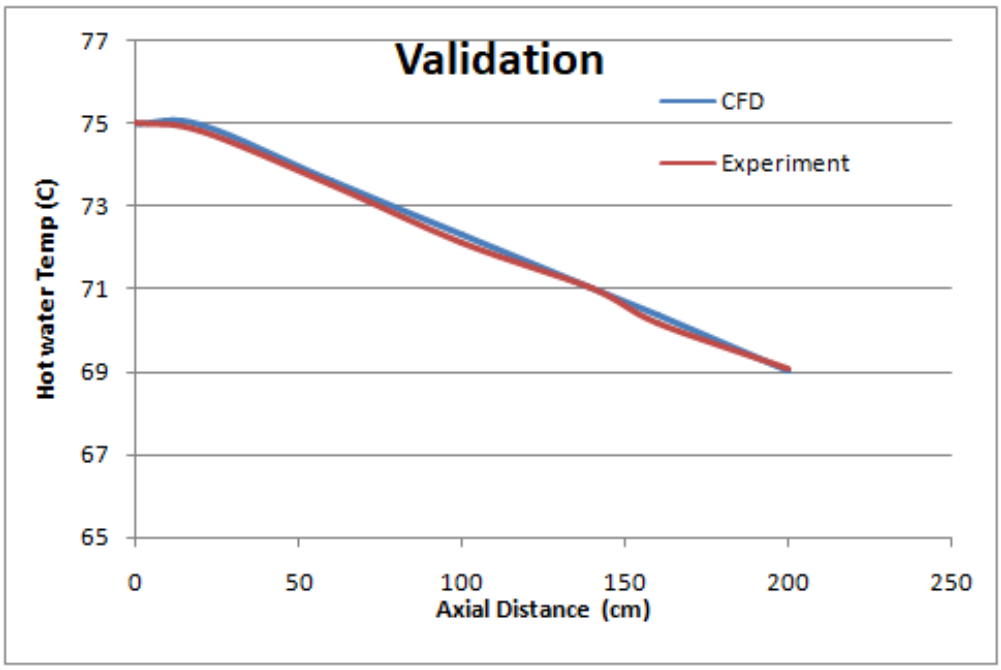


Fig 5.1 Validation of heat exchanger

CHAPTER 6

RESULTS AND DISCUSSION

In the next phase of the work, a Porcupine fins was used for the heat exchanger. The temperature distributions for heat exchanger without porcupine fins and heat exchanger equipped with porcupine fins are compared in Figure 5.1 and Figure 5.2. The pressure contours for both cases are compared in Figure 5.3 and Figure 5.4.

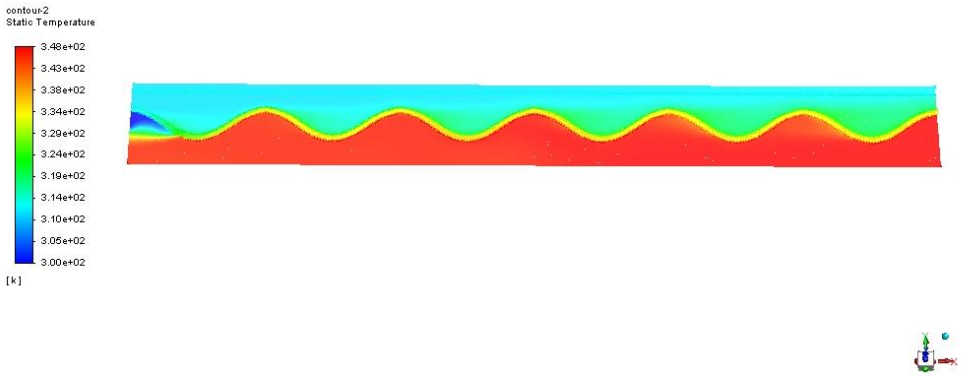


Fig 6.1 temperature contour of heat exchanger without porcupine fins

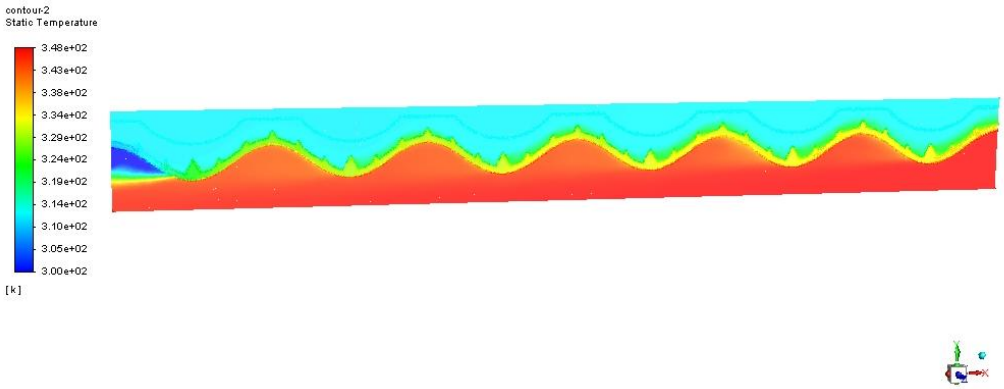


Fig 6.2 temperature contour of heat exchanger with porcupine fins

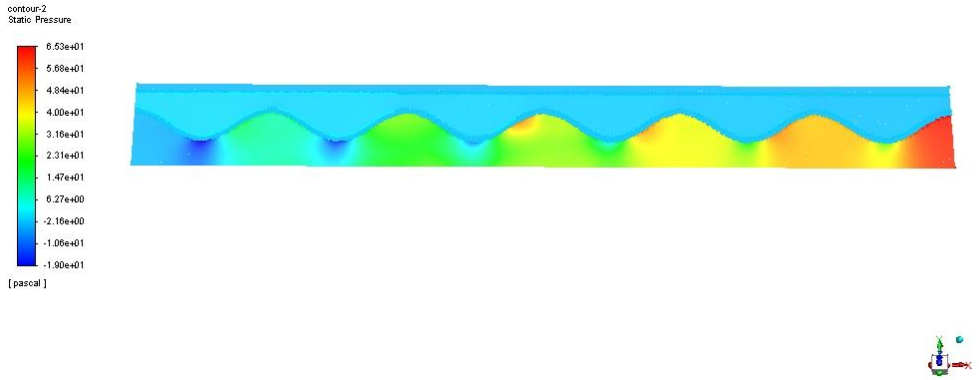


Fig 6.3 pressure contour of heat exchanger without porcupine fins

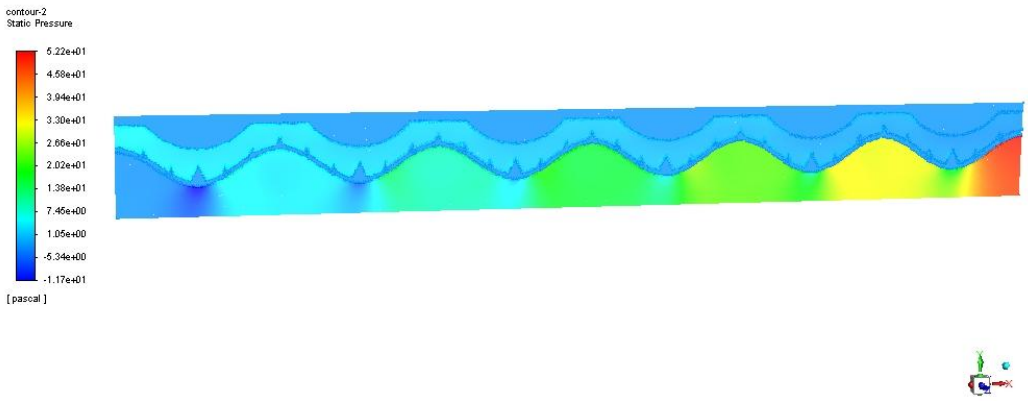


Fig 6.4 pressure contour of heat exchanger with porcupine fins

The comparisons of the temperature and pressure contours have been made at a Reynolds number of 5000. It can be observed from the comparison of temperature distributions that the hot water outlet temperature is reduced when the porcupine fins are used (337K) than when heat exchanger without fins was used (343K). This indicates that the rate of heat transfer from the hot water is increased for the porcupine fins than when the normal heat exchanger was used. Consequently, the Nusselt number is also higher when the Porcupine fins are used. It can be observed from the comparison of pressure contours that the pressure drop from inlet to outlet for hot water is significantly for higher (111 Pa) compared to the heat exchanger without fins(63 Pa). This means that the friction factor is increased when porcupine fins is used compared to the normal heat exchanger without fins for the same Reynolds number.

6.1. NUSSELT NUMBER

The comparison of Nusselt number for the heat exchanger with porcupine fins (with Al_20_3-water nanofluid) to the exchanger with porcupine fins (with water) to normal heat exchanger is as shown in Figure 6.5. The variation of Nusselt number is almost linear with increasing Reynolds number. A maximum increase of 35.44% was observed in Nusselt number for the porcupine fins (with Al_20_3-water nanofluid) compared to the other two setup. The increase was observed at a Reynolds number of 7000. The Nusselt number increase indicates a heat transfer augmentation when the porcupine fins is introduced into the outer surface of inner pipe.

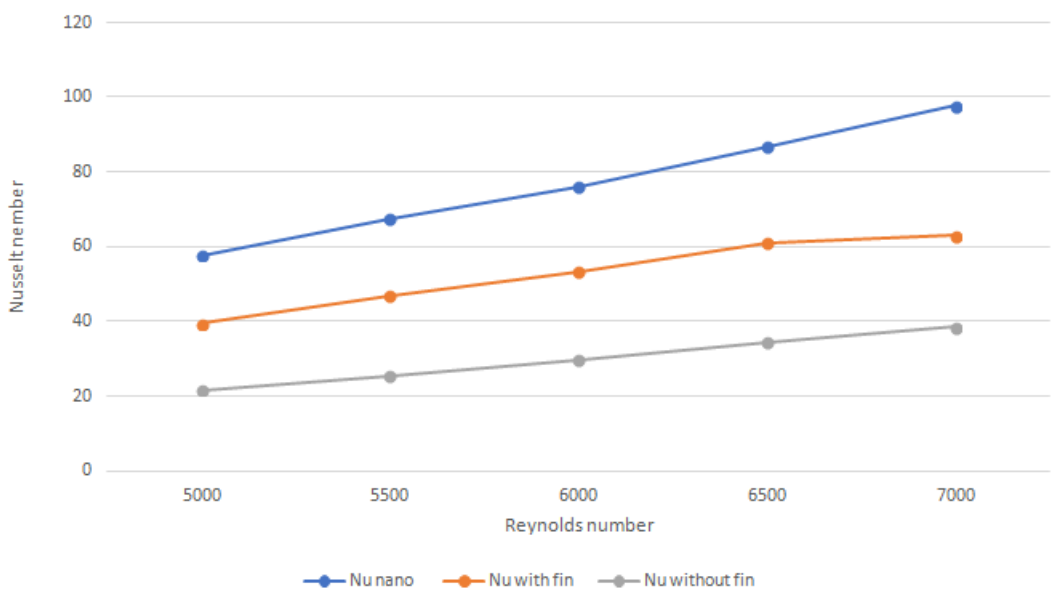


Fig 6.5 Comparison of Nusselt number

6.2. FRICTION FACTOR

The comparison of friction factor for the heat exchanger with porcupine fins (with Al_20_3-water nanofluid) to the exchanger with porcupine fins (with water) to normal heat exchanger is shown in Figure 6.6. The friction factor decreases with increasing Reynolds number in both cases. At all Reynolds numbers, the friction factor for the heat exchanger with porcupine fins is higher than that obtained for a normal heat exchanger. A maximum increase of 38.80% was observed compared to normal heat exchanger at a Reynolds number of 5000.The increase in friction factor implies that, for the porcupine fins, the pumping power required to overcome pipe friction is significantly higher than for the normal heat exchanger.

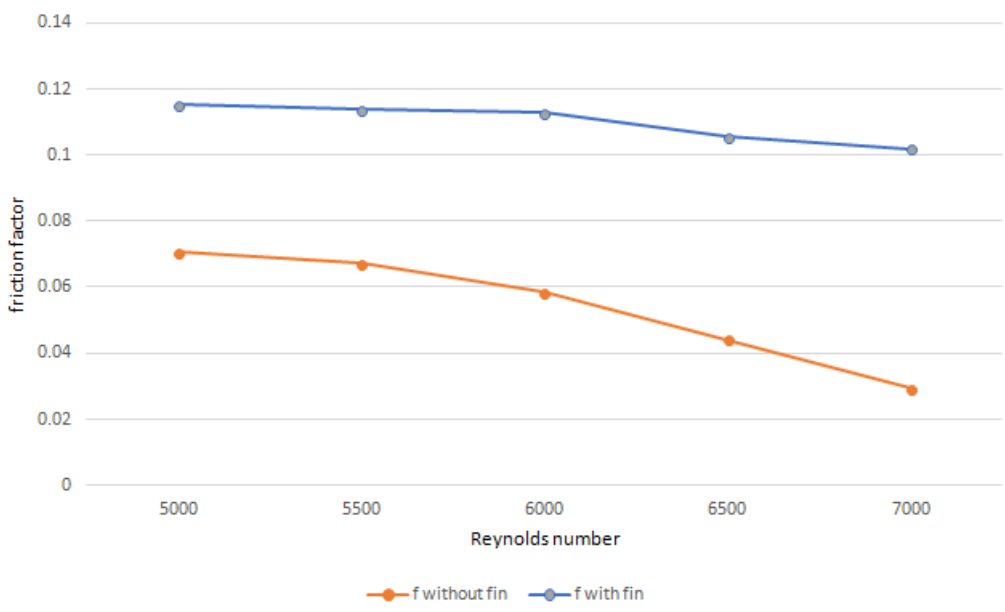


Fig 6.6 Comparison of friction factor

6.3 THERMAL ENHANCEMENT FACTOR

The thermal performance factor indicates whether the porcupine fins in heat exchanger is beneficial for use as a heat transfer enhancement device. It is desirable to have as high a Nusselt number as possible and as low a friction factor as possible.. The comparison of thermal enhancement factor for heat exchanger with porcupine fins (with Al_20_3-water nanofluid) to the exchanger with porcupine fins (with water) to normal heat exchanger is shown in Figure 6.7. A maximum increase in thermal enhancement factor of 29.96% was observed at a Reynolds number of 5000. The thermal enhancement factor decreases with increasing Reynolds number. However a 1.359 thermal enhancement factor is significant and is inferred that for the heat exchanger with porcupine fins (with Al_20_3-water nanofluid) to the heat exchanger with porcupine fins (with water) has better heat transfer and fluid flow characteristics than the normal heat exchanger.

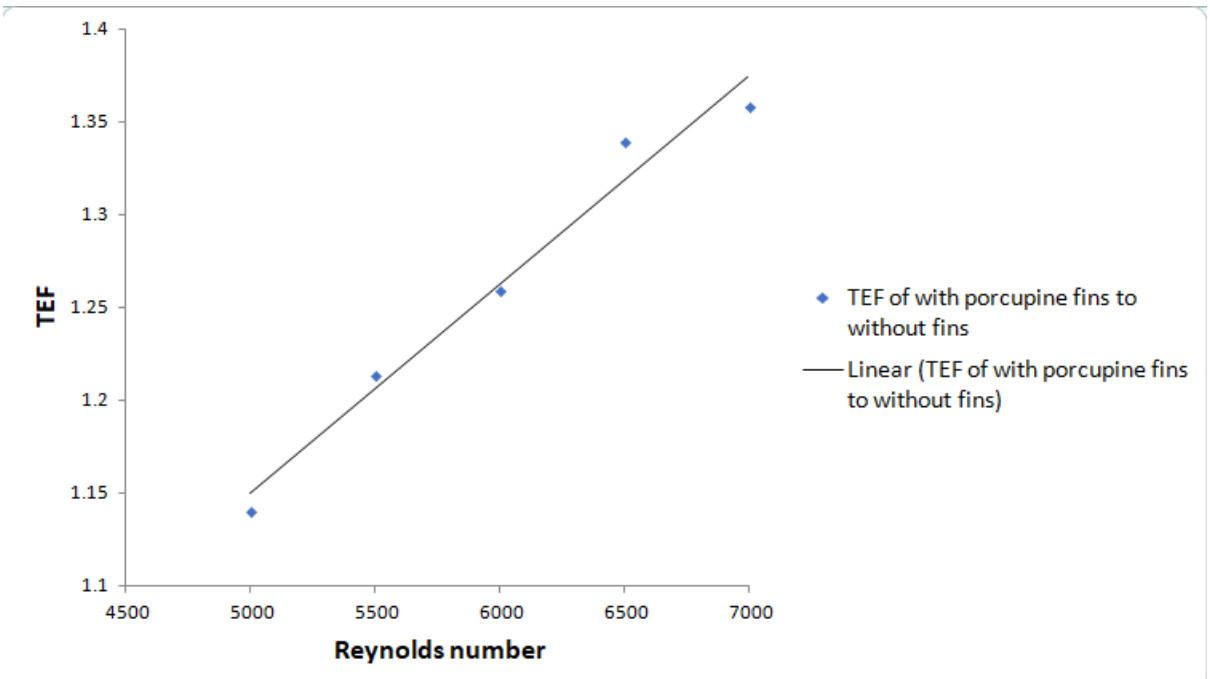


Fig 6.7 Comparison of TEF

6.4 EFFECTIVENESS

The effectiveness of a heat exchanger is defined as the ratio of the actual heat transfer to the maximum possible heat transfer. From graph, it is observed that effectiveness increases with increase in Reynolds number. Effectiveness is found to be higher when Al_20_3-water nanofluid was used, when compared to using water.

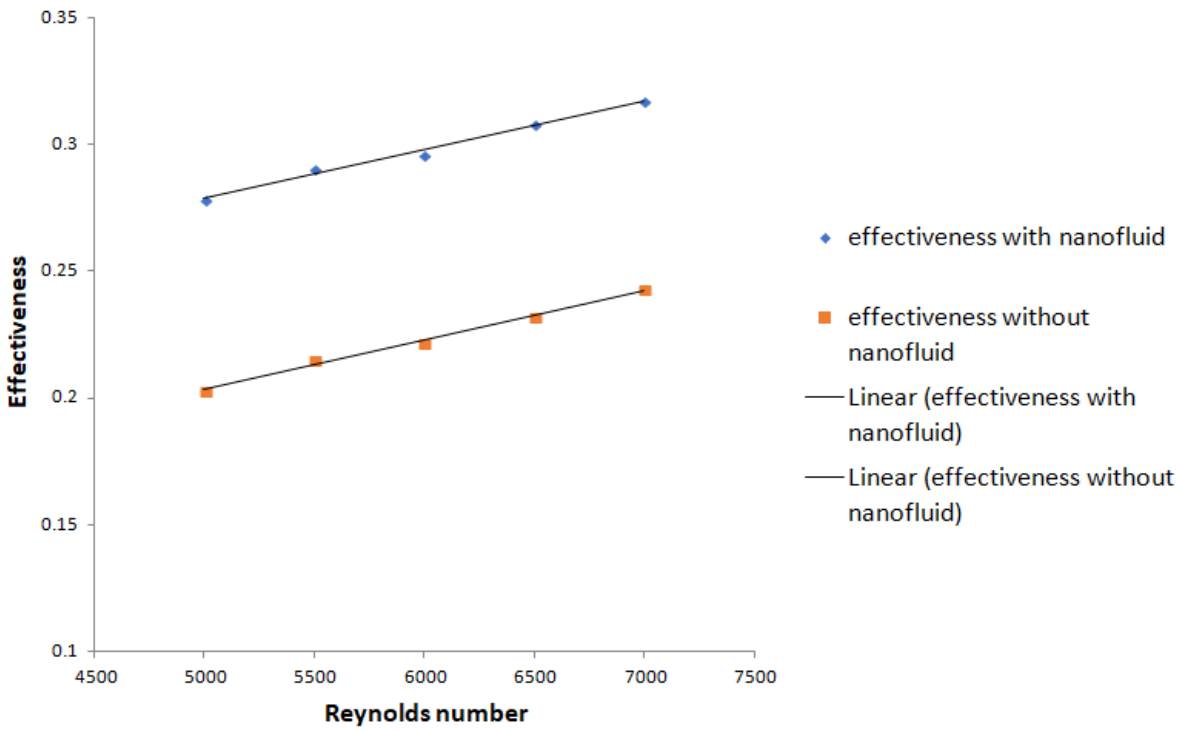


Fig 6.8 Comparison of effectiveness

CHAPTER 7

CONCLUSION

The fluid flow and heat transfer properties of a heat exchanger equipped with the heat exchanger with porcupine fins have been analyzed. Following conclusions can be drawn on the basis of the study:

- A maximum increase of 35.44% was observed in Nusselt number for the heat exchanger with porcupine fins (with Al_20_3-water nanofluid) to the exchanger with porcupine fins (with water) than the normal heat exchanger.
- A maximum decrease of 38.80 % was observed in friction factor for the heat exchanger with porcupine fins (with Al_20_3-water nanofluid) to the exchanger with porcupine fins (with water) than the normal heat exchanger.
- A maximum thermal enhancement factor of 1.359 was observed at a Reynolds number of 7000 for the heat exchanger with porcupine fins to that of the normal heat exchanger.
- A maximum effectiveness of 0.3177 was observed at Reynolds number of 7000 for heat exchanger using Al_20_3-water nanofluid while comparing heat exchanger which is using water as cold fluid.

REFERENCES

1. Chittatosh C. Bhav (2017), “A Computational Study of the Heat Transfer Characteristics of Offset-Strip Fin Cores”, Material science
2. https://imprint-india.org/knowledge-portal-6362-next-generation-heat-exchangers-design-using-additive-manufacturing-and-shape-optimization.
3. Rashmi Sahoo (2020) “Experimental study on the viscosity of hybrid nanofluid and development of a new correlation” Heat and Mass Transfer 56(7)
4. Hooman Yarmand et al. (2014) "Numercal Investigation of Heat Transfer Enhancement in a Rectangular Heated Pipe for Turbulent Nanofluid" Hindawi Publishing Corporation, Scientific World Journal., Article ID 369593, 9 pages.
5. A Azari et al (2014) " CFD and experimental investigation on the heat transfer characteristics of alumina nanofluids under the laminar flow regime, 'Brazilian Journal of Chemical Engineering' Vol. 31, No. 02, pp. 469 - 481
6. Faris Mohammed Ali et al (2011), "Study of the effect of particles size and volume fraction concentration on the thermal conductivity and thermal diffusivity of Al_2O_3 nanofluids" , "International Journal of Physical Sciences' Vol. 8(28), pp. 1442-1457
7. Azad, A.K., Saha, S.K., and Saha, S.C. (2014). Heat transfer enhancement in a plate-fin heat exchanger using triangular fins. International Journal of Heat and Mass Transfer, 78, 1125-1132.
8. Wang, Y., Zhang, Q., Yu, X., Liu, J., and Wang, X. (2019). Effects of fin density on heat transfer in a shell-and-tube heat exchanger. Energy Procedia, 158, 1783-1788.
9. Kabeel, A.E., Al-Sulaiman, F.A., and Al-Rashed, A.A. (2019). Experimental investigation of heat transfer enhancement in a shell-and-tube heat exchanger using different fin materials. Applied Thermal Engineering, 146, 281-290.

10. B.Sravana Keerthi, G.Venkatesh, Vishala (2017).Heat Transfer Enhancement Using Triangular Fin in Heat Exchanger with Nano Fluids, International Journal of Research,vol-04, issue 06
11. S Basavarajappa, G Manavendra, S B Prakash (2019). A review on performance study of finned tube heat exchanger, International Conference on Thermo-fluids and Energy Systems, 1473 (2020) 012030
12. Vinous M. Hameed, Sarah M. Awad (2021). Experimental investigation of double pipe heat exchanger temperature distribution, International Journal of Multidisciplinary Sciences and Advanced Technology Vol 2 No (10–17)